



Computer Science Department

Software Engineering

Reference Examination

2025/2026 Academic Year

Instructions: Read the case study carefully and answer all questions. Sketches may be drawn manually where required.

Case Study: Student Registration System for the University of Makeni

The University of Makeni (UNIMAK) currently manages student registration using a semi-manual process involving paper forms, spreadsheets, and multiple departmental approvals. Students often experience long queues during registration periods, delayed approval of courses, difficulty accessing academic records, and frequent data inconsistencies.

The university administration has decided to develop an integrated **Student Registration System (SRS)** to automate and improve the registration process.

The proposed system should allow students to:

- Create and maintain their student profile.
- Log into the university portal securely.
- View available courses for their program and semester.
- Register and add/drop courses during approved registration periods.
- View registration status and approval updates.
- Print registration receipts and course forms.
- Access registration history.

Academic staff and administrators should be able to:

- Create and update course offerings.
- Define registration periods.
- Approve or reject course registrations.
- Monitor student enrollment.
- Generate reports for departments and management.

The Finance Office should:

- Verify student payment status.
- Approve financially cleared students.
- Generate payment summaries.

The IT Directorate should:

- Manage user accounts and permissions.
- Monitor system performance.
- Backup and recover system data.

The university expects the system to:

- Support at least **15,000 active users**.
- Be accessible through mobile phones and computers.
- Operate continuously during registration periods.
- Protect confidential student information.
- Integrate with future examination and learning management systems.

Questions:

1. Identify and write **at least TEN (10) functional requirements** for the proposed Student Registration System. Present your answer in a tabular format: **(20 Marks)**
2. Identify and write **at least Ten (10) non-functional requirements** for the system under categories such as: Performance, Security, Availability, Reliability and Scalability: **(20 Marks)**
3. Identify all major **actors** interacting with the system. **(10 Marks)**
4. Draw a **Use Case Diagram** for the Student Registration System showing: Actors, System boundary, Main use cases and Relationships between actors and use cases: **(20 Marks)**
5. Design and sketch a **high-level System Architecture Diagram** for the Student Registration System. **(30 Marks)**